A Laboratory Manual for

Advance Java Programming (3160707)

B.E. Semester 6 (Computer)

Directorate of Technical Education, Gandhinagar, Gujarat

Government Engineering College,
Bhavnagar Certificate

This is to certify that Mr./Ms. **Maulik Kumar Pandor** Enrollment No. **230210107038** of B.E. Semester **VI**, Computer Engineering of this Institute (Institute Code: 021) has satisfactorily completed the Practical / Tutorial work for the subject **Advance Java Programming (3160707)** for the academic year 2025-26.

Place: GEC, Bhavnagar
Date:

Sign of Faculty member **Head of the Department**

Preface

Main motto of any laboratory/practical/field work is for enhancing required skills as well as creating ability amongst students to solve real time problem by developing relevant competencies in psychomotor domain. By keeping in view, GTU has designed competency focused outcome-based curriculum for engineering degree programs where sufficient weightage is given to practical work. It shows importance of enhancement of skills amongst the students and it pays attention to utilize every second of time allotted for practical amongst students, instructors and faculty members to achieve relevant outcomes by performing the experiments rather than having merely study type experiments. It is must for effective implementation of competency focused outcome-based curriculum that every practical is keenly designed to serve as a tool to develop and enhance relevant competency required by the various industry among every student. These psychomotor skills are very difficult to develop through traditional chalk and board content delivery method in the classroom. Accordingly, this lab manual is designed to focus on the industry defined relevant outcomes, rather than old practice of conducting practical to prove concept and theory.

By using this lab manual students can go through the relevant theory and procedure in advance before the actual performance which creates an interest and students can have basic idea prior to performance. This in turn enhances pre-determined outcomes amongst students. Each experiment in this manual begins with competency, industry relevant skills, course outcomes as well as practical outcomes (objectives). The students will also achieve safety and necessary precautions to be taken while performing practical.

This manual also provides guidelines to faculty members to facilitate student centric lab activities through each experiment by arranging and managing necessary resources in order that the students follow the procedures with required safety and necessary precautions to achieve the outcomes. It also gives an idea that how students will be assessed by providing rubrics.

Advance Java Programming is an Elective course. It covers various java based technologies like socket, JDBC, Servlet and JSP which help students to develop enterprise application. Web application based on Java uses Servlet, JSP, JSF etc. To store the data database connectivity and database JDBC component is needed. Networking components are needed to transfer data over network. Model-View-Controller (MVC) architecture gives flexibility and makes the web applications loosely coupled.

Utmost care has been taken while preparing this lab manual however always there is chances of improvement. Therefore, we welcome constructive suggestions for improvement and removal of errors if any.

DTE's Vision

- To provide globally competitive technical education
- Remove geographical imbalances and inconsistencies
- Develop student friendly resources with a special focus on girls' education and support to weaker sections
- Develop programs relevant to industry and create a vibrant pool of technical professionals

Institute's Vision

- To create an ecosystem for proliferation of socially responsible and technically sound engineers, innovators and entrepreneurs.

Institute's Mission

- To develop state-of-the-art laboratories and well-equipped academic infrastructure.
- To motivate faculty and staff for qualification up-gradation, and enhancement of subject knowledge.
- To promote research, innovation and real-life problem-solving skills.
- To strengthen linkages with industries, academic and research organizations.
- To reinforce concern for sustainability, natural resource conservation and social responsibility.

Department's Vision

- To create an environment for providing value-based education in Computer Engineering through innovation, team work and ethical practices.

Department's Mission

- To produce computer engineering graduates according to the needs of industry, government, society and scientific community.
- To develop state of the art computing facilities and academic infrastructure.
- To develop partnership with industries, government agencies and R & D organizations for knowledge sharing and overall development of faculties and students.
- To solve industrial, governance and societal issues by applying computing techniques.
- To create environment for research and entrepreneurship.

Programme Outcomes (POs)

1. **Engineering knowledge:** Apply the knowledge of mathematics, science,engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyzecomplex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineeringproblems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledgeand research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques,resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextualknowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professionalengineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics andresponsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as amember or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activitieswith the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understandingof the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and abilityto engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

- Sound knowledge of fundamentals of computer science and engineering including software and hardware.
- Develop the software using sound software engineering principles having web based/mobile based interface.
- Use various tools and technology supporting modern software frameworks for solving problems having large volume of data in the domain of data science and machine learning.

Program Educational Objectives (PEOs)

- Possess technical competence in solving real life problems related to Computing.
- Acquire good analysis, design, development, implementation and testing skills to formulate simple computing solutions to the business and societal needs.
- Provide requisite skills to pursue entrepreneurship, higher studies, research, and development and imbibe high degree of professionalism in the fields of computing.
- Embrace life-long learning and remain continuously employable.
- Work and excel in a highly competence supportive, multicultural, and professional environment which abiding to the legal and ethical responsibilities.

Practical – Course Outcome matrix

Course Outcomes (COs):

CO_3160707.1 Implement Networking and Data base connectivity in Java for given application.
CO_3160707.2 Implement webpage with dynamic content and server side web application using Servlet and JSP.
CO_3160707.3 Use web application framework JSF to build user interfaces.
CO_3160707.4 Use Object Relation Mapping using Hibernate to build database dependent applications
CO_3160707.5 Apply Model-View-Controller architecture to build complex client-server applications.

Sr. No.	Objective(s) of Experiment	CO 1	CO 2	CO 3	CO 4	CO 5
1.	Create databases related to your project and query them from console based java applications.	√				
2.	Create a client application that will connect with the application mentioned in the practical 1, client application will send a search "string" to the above application and above application will send back the string with the matching results after fetching from the database.	√				
3.	Modify practical 1 by replacing the console based user interface to a web based user interface. Use a servlet.		√			
4.	Modify the practical 4 to use JSP instead of Servlet.		√			
5.	Use JSF framework to replace JSP and calculate the reduction in programming			√		
6.	Make custom tag for a component that will be able to add/view/delete/modify Records			√		
7.	Use Object Relational Mapping and based on that prepare one configuration file along with the hibernet mapping file for 1 table of the application and test its working by replacing SQL to HQL.				√	
8.	Use Hibernet framework to replace JDBC calls and calculate the reduction in programming efforts for the entire application				√	
9.	Use Spring or any other MVC architecture and implement the Interface in that architecture that supports multi-tier architecture.					√
10.	Compare and analyze the JSF with the Spring framework.			√		√

Industry Relevant Skills

The following industry relevant competency are expected to be developed in the student by undertaking the practical work of this laboratory.

1. Will be able to develop web application using Netbeans/Ecllipse IDE
2. Will be able to use MVC architecture for application development
3. Will be able to use JSF and Hibernate framework

Guidelines for Faculty members

1. Teacher should provide the guideline with demonstration of practical to the students with all features.
2. Teacher shall explain basic concepts/theory related to the experiment to the students before starting of each practical
3. Involve all the students in performance of each experiment.
4. Teacher is expected to share the skills and competencies to be developed in the students and ensure that the respective skills and competencies are developed in the students after the completion of the experimentation.
5. Teachers should give opportunity to students for hands-on experience after the demonstration.
6. Teacher may provide additional knowledge and skills to the students even though not covered in the manual but are expected from the students by concerned industry.
7. Give practical assignment and assess the performance of students based on task assigned to check whether it is as per the instructions or not.
8. Teacher is expected to refer complete curriculum of the course and follow the guidelines for implementation.

Instructions for Students

1. Students are expected to carefully listen to all the theory classes delivered by the faculty members and understand the COs, content of the course, teaching and examination scheme, skill set to be developed etc.
2. We will have a project based learning methodology for practical in Advanced Java subject.
3. Students will form a team of maximum 4 people and then give the definition of a project..
4. Students can use the web application definition of Design Engineering subject also/ They can give another definition for a web application project
5. You'll need to decide the project and submit your practical list sample list is shown in next page.
6. Student should develop a habit of submitting the experimentation work as per the schedule and s/he should be well prepared for the same.

Common Safety Instructions

Students are expected to

1) switch on the PC carefully (not to use wet hands)
2) shutdown the PC properly at the end of your Lab
3) carefully Handle the peripherals (Mouse, Keyboard, Network cable etc)
4) Use Laptop in lab after getting permission from Teacher

Sample Practical List

Project Name: Movies Review Website

Project Details: In the movie review website, the administrator will be able to enter the new movie details like movie_name, genre, language, movie_certificate details, release_date, etc. The administrator should be able to add, view, delete and modify movie details. The movie details and user details should have been stored in the database. Other users will be able to search the movie and fetch the details about a particular movie. They should be able to comment and review the movie. For review, the users will be entering a positive integer number between 0 to 10. For comment, they can write texts.

Sr. No.	Objective(s) of Experiment
1	In java console application, provide the user with following options: add movie, view movie, modify movie details, delete movie details. On entering a particular option on the console by user, perform appropriate operation and display the success/failure or any output on your console.
2	Make the practical 1 application as server for sending/receiving strings. Create a client application that will send a search movie string to the server application and server will replay with the matching movie name “string” back to the client application after fetching from the database.
3	Modify practical 1 by replacing the console based user interface to a web based user interface. Use a servlet.
4	Modify the practical 4 to use JSP instead of Servlet.
5	Use JSF framework to replace JSP and calculate the reduction in programming
6	Make custom tag for a component that will be able to add/view/delete/modify Records
7	Use Object Relational Mapping and based on that prepare one configuration file along with the hibernet mapping file for 1 table of the application and test its working by replacing SQL to HQL.
8	Use Hibernet framework to replace JDBC calls and calculate the reduction in programming efforts for the entire application
9	Use Spring or any other MVC architecture and implement the Interface in that architecture that supports multi-tier architecture.
10	Compare and analyze the JSF with the Spring/any other framework.

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(Progressive Assessment Sheet)

Sr. No.	Objective(s) of Experiment	Page No.	Date of performance	Date of submission	Assessment Marks	Sign. of Teacher with date	Remarks
1.	In java console application, provide the user with following options: add movie, view movie, modify movie details, delete movie details. On entering a particular option on the console by user, perform appropriate operation and display the success/failure or any output on your console.	1	20/01	27/01			
2.	Make the practical 1 application as server for sending/receiving strings. Create a client application that will send a search movie string to the server application and server will replay with the matching movie name "string" back to the client application after fetching from the database.	7	27/01	03/02			
3.	Modify practical 1 by replacing the console based user interface to a web based user interface. Use a servlet.	12	03/02	10/02			
4.	Modify the practical 4 to use JSP instead of Servlet.	17	10/02	17/02			
5.	Use JSF framework to replace JSP and calculate the reduction in programming	22	17/02	24/02			
6.	Make custom tag for a component that will be able to add/view/delete/modify Records	30	24/02	10/03			
7.	Use Object Relational Mapping and based on that prepare one configuration file along with the hibernet mapping file for 1 table of the application and test its working by replacing SQL to HQL.	34	10/03	17/03			
8.	Use Hibernet framework to replace JDBC calls and calculate the reduction in programming efforts for the entire application	39	17/03	24/03			
9.	Use Spring or any other MVC architecture and implement the Interface in that architecture that supports multi-tier architecture.	44	24/03	07/04			
10.	Compare and analyze the JSF with the Spring/any other framework.	52	07/04	21/04			
Total							